A Serious Game for Flood Mitigation in K-12 and Public Education

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Figure 1: Snapshot from the flood game (early development stage)

ABSTRACT

This project aims to develop and evaluate a web-based serious game geared towards educating K-12 and college students, and public on flood prevention and mitigation strategies, such that they are more informed about the implications of future floods. A web-based interactive gaming environment will be developed that allows players to experiment with different flood mitigation strategies for a real-world location of their choice. This immersive, repeatable, and engaging experience will allow students and public to comprehend the consequences of individual measures, build a conceptual understanding of the benefits of mitigation actions, and examine how floods may occur in their communities.

CCS CONCEPTS

• **Information Systems**; • **Computing methodologies** → **Modelling and Simulation**; **Computer Graphics**; • **Applied Computing** → **Environmental Sciences**;

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KEYWORDS

serious gaming, socio-hydrology, computer graphics, virtual reality, disaster mitigation

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1 INTRODUCTION

The number and devastating impacts of natural disasters have grown significantly world-wide. Recent studies emphasize the importance of public awareness and training of first responders in disaster preparedness and response activities [Kapucu 2008]. However, the strategy of simply conveying the potential disaster risks in communities is not a complete solution. The likelihood and potential consequences of rare extreme events are likely to be underestimated by the public [Green et al. 1991]. Developments in serious games have shown the capacity of the medium to fulfill this need. Serious games utilize interactive and gamified story-telling mediums for education. These learning styles have been shown to have a positive effect on both student engagement as well as performance and attitude [Garneli et al. 2016]. Furthermore, digital games gained increased attention as an educational tool in the last two decades as demonstrated by the United Nations Environmental Program initiative [Patterson and Barratt 2019] for sustainability education. Current uses of serious games in environmental sciences focuses on a well-informed and technical audience and required a significant amount of training before gameplay. Thus, a gap exists in the literature to educate students and non-technical members of a community on flood mitigation with serious gaming.

This project presents a web-based serious game geared towards educating K-12 and college students, and public on flood prevention and mitigation strategies, such that they are more informed about the implications of future flooding events.A web-based interactive gaming environment is designed and implemented to allow players to experiment with different flood mitigation strategies for a real-world location of their choice. As part of the gameplay, the user will be presented with a community under risk of flooding, and various potential mitigation and preparedness action options. The users will be faced with challenges in decision-making and will have to evaluate the trade-offs in terms of the assignment of monetary resources with respect to the societal gain. Once the decisions are finalized, a realistic flood event will be generated and visualized so that the player can examine how and in what ways the community was affected. In addition to visual inspection, the proposed game will allow interactive analysis of the economic damage and casualties. This immersive, repeatable, and engaging experience will allow students and the public to comprehend the consequences of individual measures, build a conceptual understanding of the costs and benefits of mitigation actions, and examine how floods may occur in their own towns.

2 METHODOLOGY

2.1 Data Resources

The flood game utilizes various data resources (e.g. hydrological, geographical, populational) to power the game engine retrieved from federal agencies and organizations. High-resolution digital elevation model (DEM) data will be used to generate the terrain for the selected location. Methodologies for estimating property-specific damage values were developed based on HAZUS dataset as well as the data collected from the tax assessors, where available [Yildirim and Demir 2019]

2.2 Gameplay

The user will start the game by choosing a location of interest. After game environment initialization and scene construction, the user will be able to examine the city in terms of its terrain, critical infrastructure (e.g. hospital, library, schools, emergency centers, water treatment plant), vulnerable population, and economic risk. The user will be presented with a predefined budget to spend on various mitigation options within a limited time frame to increase the resilience of the community to flood events. Each mitigation option will be available with supporting information including its description, cost, and mitigation effect. Once the time allocated for preparedness is completed, the game will simulate a flood event (e.g. 100-year, 200-year, 500-year). Then, the player can examine the effected buildings, fatalities, economic damage, and agricultural and societal impact.

2.3 Software Design

The presented flood game will be implemented as a web application with main functionalities and visualizations working on the client-side for performance and efficiency. The web framework will consist of three main components (i.e. modules); Level Generation, Game Engine, and Flood Simulation. The Level Generation module will be responsible for allowing the player to select an area on the map, which will then be used to generate a gaming environment consisting of tiles (e.g. water, building, road). The Game Engine module will handle the gaming dynamics, interaction methods, mitigation strategies and effects, and extreme event generation and simulation. Finally, the Flood Simulation module will calculate the economic damages, fatalities, effected buildings, and present a detailed report that can be used to understand the impact of floods on communities as well as the benefits of preparedness.

The 3D gaming environment is developed using Three.JS with low polygon graphics. It's unique web-based design allows game play via various devices including touch-screen tables, mobile phones, and augmented and virtual reality devices.

3 SIGNIFICANCE

Studies suggest that building a conceptual understanding of mitigation strategies results in a significantly higher probability for individuals to take action and responsibilities in their communities in response to natural disasters. The proposed game enables this vision by letting the members of communities be more informed on flood risk management, understand the potential damages and casualties caused by the lack of preparedness. A major motivation to use engaging graphics and games in disaster domain is the effective education of hydrological processes, natural disasters, and social responsibilities to K-12 students. The proposed flood game can allow students to experiment with flooding scenarios for their communities to see the outcomes that were impossible to be reproduced in real life. Due to the flood game's utilization of real-world data for topography and hydrology, it has the potential to build a conceptual understanding for students in regard to extreme hydrological events.

The main goal of proposed serious game is to increase flood resilience in communities via education. It can be exhibited on a touch screen table display in museums and various outreach events for hands-on exploration by students and the public.

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